

SE Lab 3 – Architecture Selection & Justification

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Architecture Selection: We chose **Microservices Architecture** for the Remote Patient Vitals Alert & Monitoring System.

Style	Pros	Cons	Verdict
Layered	Clear separation of concerns; easy to understand	Tight coupling; hard to scale one part (ingestion)	Not chosen
Client-Server	Central control; simple deployment	Single point of failure; scalability bottleneck	Rejected
Microservices	Fault isolation; independent scaling and deployment	Operational complexity; network latency	Selected

Architectural Choice

Microservices. The system is split into independently deployable components – Telemetry Ingestion, Threshold Evaluation, Alert Escalation, Notification, Caregiver Dashboard and a Vitals Data Store – that communicate only through defined interfaces (ITelemetryStream, IVitalsEvaluation, IAlertRaise, IAlertAck, INotificationDispatch, IVitalsStorage, IVitalsQuery, IDashboardAPI, ISOSTrigger).

Two Reasons

- **Fault isolation for life-critical alerting.** The scenario demands 99.99% uptime and delivery of a critical alert within 30 seconds even if a channel fails (NFR001, NFR002). With microservices, a crash or slowdown in the Caregiver Dashboard or the data store queries cannot stop Alert Escalation or Notification. A single Client-Server backend would make one failure a single point of failure for a post-operative patient.
- **Independent scaling of the telemetry workload.** Ingestion must handle at least 500 concurrent continuous vitals streams (NFR001) and flag anomalies within 2 seconds (FR001, FR002). Ingestion and Threshold Evaluation can be scaled out on their own, while low-traffic components such as the Dashboard stay small – something a layered monolith cannot do.

Security Advantage

Each service exposes only one narrow interface, so a compromised component has a limited blast radius. Patient health data (HIPAA-class) is confined to Ingestion, Evaluation and the Data Store; the Notification Service receives only a minimal alert payload. All service-to-service traffic uses TLS 1.2+ and the Data Store is encrypted with AES-256 (NFR002), and the Dashboard Service authenticates every caregiver before acknowledgement or viewing vitals (FR004).

Performance Benefit

Asynchronous, event-driven messaging between Ingestion → Evaluation → Escalation lets each stage work in parallel without blocking, so a threshold breach reaches the caregiver within the 2-second target. The Emergency SOS path (ISOSTrigger) goes straight to Alert Escalation and bypasses threshold evaluation, meeting FR005. Trade-off: network latency and operational complexity are accepted and handled with a message queue, health checks and container orchestration.